INCH-POUND
MIL-F-22606C(SH)
11 August 1992
SUPERSEDING
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12 June 1964
(See 6.6)

MILITARY SPECIFICATION

FLASK COMPRESSED GAS AND END PLUGS FOR AIR, OXYGEN AND NITROGEN

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers seamless steel flasks with integral forged heads and various composition end plugs for storing compressed gases (air, oxygen, nitrogen, helium, or helium-oxygen mixtures).

1.2 Classification. Flasks and end plugs are of the following types, classes, and services as specified (see 6.2).

Types:

- Type GF - General service flask, straight design (see figure 1).
- Type GFS - General service flask, skirted, straight design (see figure 1).
- Type CD - General service flask, curved design (see figure 1).
- Type CDS - General service flask, skirted, curved design (see figure 1).
- Type SF - Air moisture separator flask (see figure 2).
- Type IF - Impulse flask (see figure 3).

Classes:

- Class 600 - 600 pounds per square inch (psi) maximum working pressure.
- Class 3000 - 3000 psi maximum working pressure.
- Class 5000 - 5000 psi maximum working pressure.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 05Q42, 2531 National Center Bldg 3, Washington, DC 20362-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Services:
Service A - Compressed air.
Service B - Oxygen, oil free nitrogen, helium, or helium-oxygen mixtures.

PLUGS

TYPES

Type STP - Straight plug (see figure 4).
Type RLP - Right angle (90 degree) plug (see figure 5).
Type YP - "Y" (see figure 6).
Type SFP - Separator inlet outlet plug (see figure 7).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

O-S-642 - Sodium Phosphate, Tribasic, Anhydrous; Dodecahydrate; and Monohydrate; Technical.
O-T-620 - 1,1,1-Trichloroethane, Technical Inhibited (Methyl Chloroform).
BB-N-411 - Nitrogen, Technical.
QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.

MILITARY

MIL-S-16216 - Steel Plate, Alloy, Structural, High Yield Strength (HY-80 and HY-100).
MIL-S-23009 - Steel Forgings, Alloy, High Yield Strength (HY-80 and HY100).
MIL-L-24131 - Lubricant, Colloidal Graphite in Isopropanol.
MIL-R-83248 - Rubber Fluorocarbon Elastomer, High Temperature, Fluid, and Compression Set Resistant.

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STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.
FED-STD-151 - Metals; Test Methods.

MILITARY

MIL-STD-147 - Palletized Unit Loads.
MIL-STD-271 - Requirements for Nondestructive Testing Methods.
MIL-STD-278 - Welding and Casting Standard.
MIL-STD-1688 - Fabrication, Welding and Inspection of HY-80/100 Submarine Applications.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

0900-LP-003-8000 - Surface Inspection Acceptance Standards for Metals.
59551-AM-MMM-010 - Flask High Pressure Gas Dynamic Strength Test, Procedure.

(Application for copies should be addressed to the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Steel Products Manual

AISI 304 - Stainless and Heat Resisting Steels.
AISI 4130 - Cr-Mo Alloy Steel.

(Application for copies should be addressed to the American Iron and Steel Institute, 1133 15th Street NW, Washington, DC 20005.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 285 - Standard Specification for Pressure Vessel Plates, Carbon Steel, Low- and Intermediate-Tensile Strength. (DoD adopted)

A 335 - Standard Specification for Seamless Ferritic Alloy-Steel Pipe for High-Temperature Service.

A 336 - Standard Specification for Steel Forgings, Alloy, for Pressure and High-Temperature Parts. (DoD adopted)

A 572 - Standard Specification for High-Strength Low-Alloy Columbium-Vanadium Steels of Structural Quality. (DoD adopted)

E 8 - Standard Test Methods of Tension Testing of Metallic Materials. (DoD adopted)

E 10 - Standard Test Method for Brinell Hardness of Materials. (DoD adopted)

E 23 - Standard Methods for Notched Bar Impact Testing of Metallic Materials. (DoD adopted)

E 208 - Standard Test Method for Conducting Drop-Weight Test to Determine Nil-Ductility Transition Temperature of Ferritic Steels.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section VIII, Division 1.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017)

COMPRESSED GAS ASSOCIATION (CGA)

C-1 - Methods for Hydrostatic Testing of Compressed Gas Cylinders.

(Application for copies should be addressed to the Compressed Gas Association, Crystal Gateway 1, Suite 501, 1235 Jefferson Davis Highway, Arlington, VA 22202.)

STEEL STRUCTURES PAINTING COUNCIL (SSPC)

SP 10 - Near-White Blast Cleaning.

(Application for copies should be addressed to the Steel Structures Painting Council, 4400 Fifth Avenue, Pittsburgh, PA 15213.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Material.

3.1.1 Flasks. The material for flasks shall be steel in accordance with the chemistry of class F-22 of ASTM A 336 or P-22 of ASTM A 335. Material shall be heat treated as specified in 3.3.4 to achieve the mechanical properties specified in table I.

3.1.2 Skirts and bearing blocks. The material for support skirts shall be steel in accordance with the chemistry of MIL-S-23009 or MIL-S-16216. The material for bearing blocks shall be steel in accordance with the chemistry specified in ASTM A 572. Material shall be heat treated as specified in 3.3.4 to achieve the mechanical properties specified in table I.

TABLE I. Minimum mechanical properties, flasks, skirts, and bearing blocks.

Mechanical property	Flasks ASTM A 336 class F-22 ASTM A 335 class P-22	Bearing blocks ASTM A 572	Skirts MIL-S-23009 MIL-S-16216
Yield strength at 0.2 percent offset, (psi)	90,000 (min) 130,000 (max)	42,000 (min)	90,000 (min) 130,000 (max)
Tensile strength (psi)	110,000 (min)	63,000 (min)	110,000 (min)
Elongation in 2 inches, (percent)	15 (min)	21 (min)	15 (min)
Brinell hardness (Bhn)	229 (min)	-	-
Charpy V-notch at minus 70°F (foot-pounds)	-	-	35 (min)
Nil-ductility transition temperature (NDTT)	0°F max	-	-

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3.1.3 End plugs. Unless otherwise specified, end plugs (see figures 4, 5, 6, and 7) shall be fabricated from bar stock or forged from material conforming to AISI 4130. If bar stock is used, a minimum reduction ratio of 3 to 1 from solidified cross sectional area to bar cross sectional area must be provided. They shall be suitably heat treated to achieve the properties of table II. When specified (see 6.2) the plugs may be made of QQ-N-281.

TABLE II. Mechanical properties, plugs.

Mechanical properties	AISI 4130	QQ-N-281 Hot finished
Yield strength at 0.2 percent offset (psi) (min).	55,000	30,000
Tensile strength, (psi) (min).	85,000	75,000
Elongation in 2 inches, (percent) (min)	15	26

3.1.3.1 Oxygen and special flask plugs. Inlet/outlet and drain plugs for oxygen flasks shall be nickel-copper in accordance with the chemistry only of QQ-N-281, class A, hot finished, with mechanical properties conforming to table II.

3.1.4 O-rings and back-up rings. Back-up rings shall be made of polytetrafluoroethylene. O-rings shall be in accordance with type I, class 2 of MIL-R-83248. O-rings more than 36 months old at time of initial flask shipment from the manufacturing facility shall not be used. For general configuration and assembly of O-ring and back-up ring, see figure 8, view A and figure 9.

3.2 Dimensional characteristics. The dimensional characteristics of seamless flasks shall be as specified in table III. Flasks with dimensions, pressures and volumes other than specified in table III are permitted as required by the contracting activity but shall comply with all other requirements of this specification (see 6.2). Unless otherwise specified herein, the difference between the major and minor diameters shall not exceed 2 percent of the outside diameter (od) of the flask. For curved type CD and CDS flasks, the difference in out of roundness between the major and minor diameter shall not exceed 3.5 percent of the od of the body. For typical configuration of the flasks, see figures 1 and 2. Impulse flasks shall conform to the characteristics and dimensions of figure 3, (see 6.2).

3.2.1 Wall thickness. Flasks shall be manufactured to the design wall thickness specified in table III based on the minimum tensile strength specified in table I. However, a wall thickness less than that required by table III is acceptable provided that this wall thickness is equal to or greater than that specified in table IV for the actual tensile strength of material for the flask, and all other tests and inspection requirements are met.

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TABLE III. Dimensional characteristics of flasks (less end plugs).

Class and type	Minimum capacity (cubic feet)	Outside diameter 2/4/ (inches)	Overall length (plus 1/2 inch minus 0 inch) (feet-inches)	Wall thickness 3/ (min) (inches)	Weight (approx) (pounds)	Hydrostatic test pressure (psi)
Class 600						
Type IP	7.0	18	5-10-1/2	0.250	720	1000
Class 3000						
Type GF	4.0	18	4-1/2	0.561	450	5000
Type GF	6.0	18	5-4-1/2	.561	615	5000
Type GF	8.0	18	6-8-1/2	.561	800	5000
Type GF	10.0	18	7-11-1/2	.561	915	5000
Type GF	15.5	18	11-8	.561	1350	5000
Type GF	21.0	18	15-3-1/2	.561	1770	5000
Type CD	21.0	18	15-3-1/2	.561	1770	5000
Class 5000						
Type SF	1/ 0.5	6-5/8	3-11-1/2	.494	140	8330
Type GF	1/ 1.5	10-3/4	4-3-1/2	.572	305	8330
Type GF	4.0	18	4-4-1/2	.933	850	8330
Type GF	6.0	18	5-10	.933	1135	8330
Type GF	8.0	18	7-3-1/2	.933	1415	8330
Type GF	10.0	18	8-9-1/2	.933	1705	8330
Type GF	10.0	20	7-5-1/2	1.028	1715	8330
Type GF	15.5	18	12-10	0.933	2480	8330
Type CD	15.5	18	12-10	.933	2480	8330
Type GF	21.0	18	16-10-1/2	.933	3260	8330
Type CD	21.0	18	16-10-1/2	.933	3260	8330
Type GF	25.0	20	16-4-1/2	1.028	3820	8330

1/ The type SF and 1.5-cubic foot type GF flasks shall be class 5000 only and shall be used for all applications up to 5000 psi.
2/ Measure the od tolerance and out of roundness as specified in 4.3.2.
3/ Wall thickness includes a minimum corrosion allowance of 0.041-inch for class 3000 flasks and 0.0625-inch for class 5000 flasks.
4/ For flasks up to and including 18-inch od the tolerance is +1/4, -1/16 inch. For 20-inch od flasks the od tolerance is +5/16, -1/16 inch.

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TABLE IV. Minimum allowable wall thickness based upon actual tensile strength.

Actual tensile strength of material (psi)	Class 3000	Class 5000			
	18 od (inch)	6-5/8 od (inch)	10-3/4 od (inch)	18 od (inch)	20 od (inch)
112,000	0.551	0.489	0.562	0.917	1.010
114,000	.542	.483	.553	.902	.993
116,000	.534	.478	.545	.888	.997
118,000	.525	.473	.536	.874	.962
120,000	.517	.468	.528	.860	.947
122,000	.509	.463	.520	.847	.932
124,000	.502	.458	.513	.834	.918
126,000	.494	.454	.505	.822	.904
128,000	.487	.449	.498	.810	.891
130,000	.480	.445	.491	.798	.878

3.2.1.1 Wall thickness exception. In addition to the provisions of table IV, flasks may have isolated areas with a wall thickness less than that specified for a corresponding actual tensile strength of the material if the reinforcement of that isolated area meets the requirements of section UG-40 of the ASME BPV Code, Section VIII, Division 1. The minimum wall thickness in the neck to head transition area (see figure 10) shall be equal to or greater than 75 percent of the minimum design wall thickness specified in table III.

3.3 Fabrication. The body of types GF, SF, and GFS flasks shall be in the shape of a straight cylinder as shown on figures 1 and 2. The types CD and CDS flasks shall be a curved cylinder as shown on figure 1. Each end shall be hot formed to provide a neck for the flask end connections (see figure 10). Flasks may be fabricated of either seamless tubular material or pierced and drawn billets. The type SF flask may be fabricated with a steel ring shrunk onto the neck to achieve the required 4.25-inch neck od. The ring shall be of the same material as the flask and shall have a 0.002-inch minimum interference fit. Skirts and bearing blocks for types GFS and CDS flasks shall be as specified by the contracting activity (see 6.2). If an internal drain for horizontal installation is required, it shall be in accordance with figure 14 and as specified (see 6.2). The internal drain tube shall be fabricated of AISI 304.

3.3.1 Oil. Oils used in the production process during fabrication shall be of the type that will fluoresce under an ultraviolet light examination (see 4.3.3).

3.3.2 Mercury. Mercury-bearing instruments or equipment which may cause contamination shall not be used in manufacture, fabrication, assembly, or testing unless such instruments or equipment are protected by a double containment boundary. In the event of suspected mercury contamination or lack of mercury free material certifications, testing may be conducted to determine if excessive contamination is present. Surface contamination shall not exceed 0.01 milligrams per cubic meter and internal contamination shall not exceed 10 parts per million.

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3.3.3 Fabrication welds. All welding shall be in accordance with MIL-STD-278. Inspection shall be conducted as specified in 4.4.4.

3.3.4 Heat treatment. After hot forming operations have been completed, as well as any welding on the flask body for other than seal welds, the flask shall be heated to above the upper critical transformation temperature and quenched in a suitable medium and then tempered to achieve the required mechanical properties specified in table I. The minimum tempering temperature shall be 1100 degrees Fahrenheit (°F). After heat treatment, welds shall be examined as specified in 4.4.4. If a weld joint repair is made within 3/8-inch of the flask body, the flask shall again be fully heat treated, and all welds re-examined.

3.3.5 Threads. Threads shall be Unified National Threads in accordance with FED-STD-H28 or as specified (see 6.2 and figures 4, 5, 6, 7, and 10).

3.3.6 End plugs. Unless otherwise specified (see 6.2) end plugs shall be manufactured in the configurations shown on figures 4, 5, 6, and 7.

3.4 Surface preparation. Flask interiors and exteriors shall be cleaned and free of mill scale, rust, manufacturing oils and other dirt or debris that would interfere with the application of a protective coating. A recommended procedure is to steam clean, dry, grit blast, and vacuum to remove all unwanted contamination.

3.4.1 Blasting. Flask surfaces shall be abrasive blasted to a minimum of near-white finish in accordance with SSPS SP 10. The abrasive and particle size shall be selected by the manufacturer to produce an od surface profile of between 0.0015-inch and 0.0025-inch as measured by visual comparison with a reference standard. The abrasive debris shall be removed by vacuuming or other suitable means. Protection shall be provided for neck threads during grit blasting operations.

3.4.2 Contamination. The flasks, when cleaned, shall be subjected to the oil-free test of 4.3.3. Flasks showing any trace of oil contamination shall be recleaned as specified in 3.4.1. Flasks shall be recleaned until the ultraviolet light examination shows no oil contamination. If turning or rust bloom occurs, flasks must be reblasted to near-white minimum and retested with ultraviolet light.

3.4.3 Drying. The flasks shall be thoroughly dried prior to painting or inerting.

3.4.4 Plugs. Plugs shall be free of oil, grit, products of machining, and so forth. They shall be wire brushed and rinsed with detergent trisodium phosphate (O-S-642), or trichloroethane (O-T-620). Plugs shall be recleaned until ultraviolet light examination shows no oil contamination.

3.5 Surface protection. Within 24 hours after cleaning and drying, the flasks shall be given internal and external surface protection as follows. If flask is to remain uncoated for 24 hours or more, then protection against corrosion must be provided. Protection for threaded areas must also be provided during the coating process.

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3.5.1 Interior. Unless otherwise specified (see 6.2) service A flask interiors shall be coated with either an organic or inorganic zinc rich coating system with a minimum dry thickness of 0.003-inch. Service B flask interiors shall, if specified, (see 6.2) be phosphated.

3.5.2 Exterior. Unless otherwise specified (see 6.2) both service A and service B flask exteriors shall be coated with either an organic or inorganic primer system to a minimum dry film thickness of 0.002-inch and a maximum dry film thickness of 0.004-inch. Film thickness shall be determined at a minimum of five random locations.

3.6 Pressure testing.

3.6.1 Hydrostatic. After final heat treatment, but prior to final cleaning and painting, each flask shall be hydrostatically tested as specified in 4.4.3.1.

3.6.2 Plugs. Each plug, prior to flask assembly, shall be hydrostatically proof tested as specified in 4.4.3.3. Plugs which pass the hydrostatic proof test shall be marked with the letter "H" as specified in 3.7.2(d).

3.6.3 Gas leakage. Each flask after final assembly shall be gas leakage tested (see 4.4.3.2).

3.7 Marking for identification. Flasks and plugs shall be permanently marked as specified in 3.7.1 or 3.7.2 as applicable.

3.7.1 Flasks. Permanent marking for identification of flasks shall be stamped on the cylindrical portion of the neck not closer than 1/2-inch to the tangent at the root of the neck. Such markings shall be made with a round bottom low stress stamp and shall not be over 1/16-inch deep. Stamps with characters greater than 1/4-inch in height shall not be used. Flasks with skirts shall have the specified stamping applied to the skirt rather than on the flask neck. The following information shall be stamped on the neck or skirt of each flask in the order listed:

		Example of stamping
(a)	Manufacturer's initial or trademark and date of hydrostatic test, month and year.	CPI, 7-91
(b)	Specification, type, class, and service.	22606C-GF-5000-A
(c)	Nominal capacity and weight, diameter, and manufacturer's serial number.	10--1705--18--XXXX
(d)	Design min. wall thickness (inch).	0.933
(e)	Minimum wall permitted by table IV using prefix "M" (inch).	M.860
(f)	Actual measured wall thickness using prefix "A". Use only when 3.2.1 and table IV are invoked (inch).	A.902

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Identification markings used by a manufacturer during fabrication may be stamped longitudinally on the cylindrical portion of the flask prior to forging and heat treating operations. Such markings shall be made with a round bottom, low stress stamp and shall not be over 1/16-inch deep. Stamps with characters greater than 1/4-inch in height shall not be used. This identification need not be removed after final finishing operations.

3.7.2 Plugs. Permanent marking for identification of plugs shall be made by using stamping with round bottom low stress die stamps. This marking shall be applied to any suitable surface which will be exposed after assembly of the plug to the flask and shall contain, as a minimum, the following identification:

		Example of stamping
(a)	Manufacturer's initial or trademark.	CPI
(b)	Plug material.	AISI 4130
(c)	Heat no. or code for plug.	E6
(d)	Hydrostatic test designation (see 3.6.2). Place immediately after mfr's initial.	H
(e)	Part no. or drawing no. of plug.	2MP2945

3.7.3 Tagging. Each flask shall bear a weather-resistant tag reading, "This flask inerted with dry nitrogen (BB-N-411)".

3.8 Assembly. Non-welded plugs shall be assembled to the flask with O-rings, back-up rings, and retaining bands (see figure 8). Plugs shall be torqued to 250 to 400 foot-pounds. Seal welded plugs shall be torqued to 800 +/-200 foot-pounds prior to seal welding. O-rings shall not be installed into plugs that are to be seal welded. Refer to figure 13 for a typical seal weld (AISI 4130 plug to ASTM A 336, F22 alloy flask). When assembling the plug to the flask, no gap shall exist between the plug shoulder and the flask sealing surface flat. The inerting caps shall be in accordance with figure 11. Completed seal welds shall be inspected as specified in 4.4.4.1. Seal welding procedures shall be approved by contracting activities.

3.8.1 Thread coating. Threads shall be lubricated with graphite in isopropanol in accordance with MIL-L-24131.

3.9 Inerting. Flasks shall be preserved as follows: After assembly, each flask shall be evacuated to an absolute pressure of 3 to 8 inches mercury. From this partial vacuum, the flask shall be pressurized to a pressure of 5 to 25 psi with nitrogen which is in accordance with BB-N-411. The charging pressure shall be determined at the ambient temperature when the flask is charged. Inerting plugs shall be as shown on figures 11 and 12.

3.9.1 Inerting for emergency shipment. Flasks destined for emergency shipment where the carrier will not accept charged flasks shall be charged to approximately 5 psi with dry nitrogen in accordance with BB-N-411 then bled down to 0 psi to meet carrier requirements.

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3.10 Recovered materials. Unless otherwise specified herein, all equipment, material and articles incorporated in the products covered by this specification shall be new and may not be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.11 Workmanship. Flasks and plugs shall be free from dirt, manufacturing debris, and oils. Threads shall be smooth, properly formed, and free from burrs and nicks. Protective coatings shall be continuous, adherent, and free from blisters, streaks, and bare spots.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Sampling for inspection.

4.2.1 Lot definitions.

4.2.1.1 Lot-flasks. A lot (for heat treat purpose only) shall consist of flasks of the same material, diameter and wall thickness from approved heats of steel and heat treated in the same heat treatment furnace charge. An approved heat of steel is defined as one from which a test sample prepared as specified in 4.2.2 and heat treated as specified in 3.3.4 has been tested and shown to have mechanical properties equal to those specified in table I. A furnace charge is defined as flasks heat treated at the same time in a batch-type furnace or up to a maximum elapsed time of 4 hours of production in a continuous furnace.

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4.2.1.2 Lot-non-pressure boundary parts. In the case of non-pressure boundary parts which are attached by welding to the flask, a lot shall consist of parts from an approved heat of steel. An approved heat of steel is defined as one from which a test sample of the greatest material thickness supplied from the heat has been heat treated as specified in 3.3.4 and has been tested and shown to have mechanical properties equal to those specified in table I. Non-pressure boundary parts may include, but are not limited to, such items as fitting and mounting attachments (skirts, mounting blocks, body bands), and accessory attachment equipment (pressure or temperature sensor sites, and so forth).

4.2.1.3 Lot-plugs. A lot (for heat treat purpose only) shall consist of plugs of the same material, heat, method of fabrication (forged or barstock), and treated in the same heat treatment furnace charge.

4.2.2 Test samples.

4.2.2.1 Test samples representing flasks. A circular test sample (ring) at least 24 inches long shall be heat treated with each lot of flasks. The test sample shall be in the shape of a right circular cylinder with ends closed (where required) and having the same material, diameter, and wall thickness as the flasks which it represents. The sample shall be subjected to the same manufacturing processes (except end closure) and heat treatment as the flasks which it represents. The methods of quenching the test sample shall duplicate that of the flasks. If the flasks are quenched from the outside only, then the test sample shall have its ends closed; however, one end may be vented as necessary to prevent pressure buildup during heat treating. The following tests require specimens that shall be cut from the test sample. A flask from the heat treat lot may be consumed for test specimens in place of a test ring.

(a) Yield and tensile (see 4.4.2.1).
(b) Charpy (see 4.4.2.2).
(c) Drop weight (NDT) test (see 4.4.2.3).

4.2.2.2 Test samples representing non-pressure boundary parts. Test samples for use in determining lot approval of non-pressure boundary parts are to be quenched in the same orientation and same quench medium exposure as they will be during flask heat treatment. Specimen selection shall be made to obtain necessary tests. Non-pressure boundary parts which have been subjected to the lot approval procedure specified in 4.2.1.2, and which have met the mechanical property requirements of table I are not subject to additional testing with each heat treat furnace charge.

4.2.2.3 Test samples representing plugs. Test samples for use in determining lot approval of plugs shall be obtained from as heat-treated stock. If individual forgings are heat treated for plug manufacture, a treated forging from each lot will be used for a test sample. If heat treated bar stock is used for plug manufacture, a test sample shall be obtained out of a bar from each lot. Required mechanical properties for end plugs are specified in table II.

4.3 Visual and dimensional examination. The following examinations shall be performed as specified (see 6.3).

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4.3.1 Visual examination. Each flask shall be visually examined to verify conformance to this specification. Flasks shall be clean as specified in 3.4. Surface imperfections, such as round-bottom pits, scores, gouges, and so forth are not cause for rejection unless the depth of the imperfection violates the minimum design wall thickness of the flask except as modified by table IV. Such imperfections, if greater than 1/16 inch deep, shall be faired at not less than a 4 to 1 ratio into the adjacent area by grinding or machining. Tightly adhering weld spatter which is acceptable in accordance with MIL-STD-1688 shall not be cause for rejection.

4.3.1.1 Formed ends and neck examination. After the hydrostatic test specified in 4.4.3.1 is conducted, integrally formed ends and necks of each flask shall be visually examined for surface cracks. Surface discontinuities such as blemishes, scale marks, chip marks, hammer marks, folds, seams, and scale that would not be considered cracks and would not be rejectable in the cylindrical section also would not be considered cracks and would not be rejectable on the integrally formed ends and necks. Flasks submitted under the wall thickness exception of 3.2.1.1 will be accepted provided:

(a) All visible forging folds on the inside surface are ground out.
(b) Irregularities on the outside surface in the transition region, such as from chipping or hammering, are ground out.
(c) The final thickness measurement is performed after such grinding.

Flasks containing visual conditions other than those described above shall not be offered for delivery without approval of the contracting authority.

4.3.2 Dimensional examination. Flask pipe material shall be 100 percent ultrasonically inspected to determine wall thickness prior to flask fabrication. Ultrasonic inspection shall be in accordance with MIL-STD-271. This measurement shall consider reductions due to scaling and grit blasting. After grit blasting of the outside surface to remove heat treat scale, but before painting, the flask diameter shall be measured. The diameter shall be determined near the center of the flask. This measurement may be made with a wrap tape or other suitable tool. Flask ovality may be determined by micrometers, calipers or other suitable tools. Wall thickness shall be determined at all outside and inside surface ground or machined repair areas by ultrasonic thickness measuring techniques. The flask length shall be measured after final machining of the nozzle faces but before installation of the end plugs.

4.3.3 Oil-free examination. All flasks, after cleaning as specified in 3.4, shall be inspected internally by ultraviolet light (black light). If there is any evidence of fluorescence on the interior or threaded surfaces, the flask shall be recleaned (see 3.4) and resubmitted to the oil-free test. In addition, service A flasks shall be subject to the same oil free test after internal surface protection as specified in 3.5.1 is completed.

4.4 Tests. The following tests shall be performed as specified (see 6.3).

4.4.1 Chemical analysis. Prior to flask fabrication, the producing mill shall take a sample from a pipe or billet of each heat of steel to be used for flask fabrication and which shall be subjected to a chemical analysis to determine

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compliance with 3.1.1. Sample selection and testing shall conform to test method 111.2 of FED-STD-151. Failure of the chemical analysis to conform to specified material check analysis limits shall be cause for rejection of the heat of material represented.

4.4.2 Mechanical tests.

4.4.2.1 Yield and tensile (applicable to all steel grades). Four longitudinal round specimens spaced 90 degrees apart shall be cut from the test sample (see 4.2.2), and shall be tested in accordance with ASTM E 8 to determine compliance with yield, tensile, and elongation properties specified in table I. Reheat treatment is permitted in the event of a test failure. Failure to pass this test shall be cause for rejection of the lot represented.

4.4.2.2 Charpy test (applicable only for HY-80 steel). Three longitudinal Charpy V-notch test specimens shall be cut from the test sample (see 4.2.2) and tested in accordance with ASTM E 23 to determine conformance to table I. The specimens shall be selected from an upper quadrant of the test sample. The value specified in table I shall be the minimum average value of three specimens. However, no single specimen shall be below the specified minimum by more than 5 foot-pounds. Reheat treatment is permitted in the event of a test failure. Failure to pass this test shall be cause for rejection of the lot represented.

4.4.2.3 Drop weight test (applicable only for F-22 and P-22 steels). The drop weight test shall be conducted in accordance with ASTM E 208. Not less than three longitudinal specimens taken from each test sample (see 4.2.2) at locations adjacent to tensile test specimens shall be tested at a temperature of 10°F. Specimens must pass in order to constitute a valid test. This indicates an NDT temperature of at most 0°F in accordance with the method specified in ASTM E 208. The proper weight height shall be a height that assures deflection of all specimens to contact anvil stops. Where the flask wall thickness precludes the use of a 5- by 2- by 5/8-inch test specimen, a 5- by 2- by 1/2-inch test specimen may be utilized. The dimension for the deflection stop for the 1/2-inch thick specimen shall be 0.090 + 0.004/- 0-inch in lieu of 0.075 + 0.004/- 0-inch. Reheat treatment is permitted in the event of a test failure. Failure shall be cause to reject the lot represented.

4.4.2.4 Brinell hardness testing (flasks). Each flask shall be Brinell tested on the cylindrical section in accordance with ASTM E 10. Hardness so determined shall equal or exceed 229 BHN. A minimum of two readings shall be obtained on each flask, one at each end, 180 degrees apart.

4.4.3 Pressure tests.

4.4.3.1 Hydrostatic test. Each flask shall be tested to 5/3 of the design working pressure of the flask. Testing shall be either the Dynamic Pressure Test in accordance with NAVSEA S9551-AM-MMM-010 or the Water Jacket, Leveling Burette Test in accordance with CGA C-1. A deviation from a straight line on the Dynamic Pressure Test graph shall constitute failure of the test. A permanent expansion exceeding 10 percent of the total volumetric expansion during the Water Jacket test shall constitute a failure. Tap water shall be used for this test. Neither

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the water nor the flask shall be heated. The test pressure shall be held at least 5 minutes. Any flask that fails this test shall not be offered for delivery. In the event of a flask failure the manufacturer may apply for a one-time reheat treatment and retest. Application shall be made to the contracting activity.

4.4.3.2 Gas leakage test. After the hydrostatic test of 4.4.3.1 and inspection specified in 4.4.4, each flask shall be pressurized with clean, dry air or nitrogen to the design working pressure of the flask. This pressure shall be held for not less than 5 minutes. There shall be no evidence of leakage through any portion of the seal weld or end plug as determined with a leak test solution. Plug seal weld repair is permitted. Any flask that fails to comply with this test shall not be offered for delivery.

4.4.3.3 Flask plug proof test. All plugs intended for use with flasks made to this specification shall be pressurized, using clean water, to 8330 +200/-0 psi. The pressure shall be held for not less than 5 minutes, and there shall be no leakage as a result of this test.

4.4.4 Nondestructive tests.

4.4.4.1 Weld inspection. All multi-pass fabrication and attachment welds, other than seal welds, shall have a separate root pass inspection. All fabrication and attachment welds shall be inspected following flask heat treatment. Seal welds shall be inspected following completion. Weld inspection shall be by magnetic particle (MT) or liquid penetrant (PT) examination in accordance with MIL-STD-271. MT is the preferred inspection method. PT may be substituted for MT where MT is impractical. All weld repairs shall be re-examined as specified in this paragraph.

4.4.4.1.1 Weld acceptance. Weld acceptance shall be in accordance with class 1 of NAVSEA 0900-LP-003-8000.

4.4.4.2 Flask magnetic particle inspection. After the hydrostatic test has been completed, hemispherical ends and necks of each flask shall be subjected to magnetic particle examination in accordance with MIL-STD-271.

4.4.4.2.1 Flask magnetic particle acceptance. Acceptance shall be in accordance with NAVSEA 0900-LP-003-8000.

4.4.4.3 Flask ultrasonic inspection. After the hydrostatic test has been completed, each flask shall be 100 percent ultrasonically examined over the cylindrical section by shear wave in both the circumferential and axial directions, to a point 3 inches beyond the start of curvature of the integrally formed head. Flasks shall also be 100 percent inspected over the entire surface by longitudinal wave. Ultrasonic examination shall be conducted in accordance with MIL-STD-271.

4.4.4.3.1 Flask shear wave acceptance. Any flask having an indication greater than 100 percent of the calibration level shall be rejected. Any flask having an indication of between 50 and 100 percent of the calibration level, and over 3 inches in length, shall be rejected unless approved by NAVSEA or the agency concerned. Other ultrasonic indications over 25 percent of the calibration level shall also be submitted for information to NAVSEA or the agency concerned as part of the test report.

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4.4.4.3.2 Flask longitudinal wave acceptance. Any flask showing a 100 percent loss of back reflection shall be rejected. Flasks showing a loss of back reflection of between 50 and 100 percent, which is linearly disposed, and which totals 6-inches or more in 12-inches of flask length, shall be rejected. Areas of flasks showing a loss of back reflection between 50 and 100 percent, that are greater than 3-inches in length, shall be noted in the report. Any discontinuity signal accompanied by a loss of back reflection greater than 25 percent shall also be reported.

4.5 Retests.

4.5.1 Mechanical retesting. The manufacturer may reheat treat the lot, and repeat the specified tests, if the specimens fail the tests specified in 4.4.2.

4.5.2 Pressure tests. Flasks failing the hydrostatic test of 4.4.3.1 may, upon request (see 4.4.3.1), be reheat treated and tested as specified in 4.4.3.1 and 4.4.4. Flasks failing the gas leakage test shall have the O-ring seal or defective seal weld area repaired and then be retested as specified in 4.4.3.2.

4.5.3 Nondestructive and visual inspection. Surface imperfections detected by the methods specified in 4.3 or 4.4.4 which would require rejection but can be made acceptable by grinding, machining or cleaning, may be so treated and then reexamined by the same method as disclosed the imperfections.

4.6 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Preservation. Not applicable.

5.2 Packing. Packing shall be levels A, B, or C as specified (see 6.2).

5.2.1 Levels A and B. Flasks inerted with nitrogen (see 3.9) and of like types may be strapped to pallets in such a manner to permit the use of lifting devices and material handling equipment. MIL-STD-147 provides guidance for palletized load but is not mandatory. Large curved and straight flasks may be shipped without pallets but must be blocked and braced in such a way to prevent shifting during movement.

5.2.2 Level C. Packing shall be accomplished in a manner which will ensure acceptance by common carrier and shall afford protection against physical and mechanical damage from handling during direct shipment from the supply source to the using activity. The method of packing shall conform to the carrier regulations as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required (see 6.2), marking of flasks and palletized loads shall be composed of the following information:

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(a) NSN/NATO stock number. When shown, the stock number shall include the appropriate spaces or dashes and any prefix or suffix shown in the contract/requisition.
(b) CAGE code (previously FSCM). When the stock number is not available or when required in the contract, the Commercial and Government Entity (CAGE) code applicable to the design activity shall be shown followed on the next line by the CAGE code of the actual manufacturer or supplier if different. The letters "CAGE" and "CAGE/MFR" shall be used to identify this information.
(c) Item description.
(d) Quantity and unit of issue.
(e) Levels of protection and date packed (for example, A/B 12/91 indicates level A preservation, level B packing, and 12/91 as the earliest pack date (month and year) for the unit pack). Unit packs used as exterior containers (such as, bags, drums, and so forth) shall have only the level of pack and date applied.
(f) Gross weight and cube.
(g) Contract number or purchase order number.
(h) Name and address (including zip code) of the contractor. When the supplies are shipped from a subcontractor, only the name and address of the prime contractor shall be used.
(i) Ship to address. This may be made by stenciling, silk-screening, labels or tags. Military Label DD Form 1387 is preferred.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. These flasks are intended for, but not limited to, shipboard use in the storage of high-pressure air, oxygen, nitrogen, helium, or helium/oxygen mixtures.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Flask type (see 1.2).
(c) Flask class (see 1.2).
(d) Flask service (see 1.2).
(e) Plug type (see 1.2 and 3.3.6).
(f) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(g) Plug material (see 3.1.3).
(h) Special sizes and configurations (see 3.2).
(i) Skirt and bearing block details if required (see 3.3).
(j) Internal drain and details if required (see 3.3).
(k) Type and size of threads (see 3.3.5).
(l) Internal preservation if other than specified (see 3.5.1).
(m) External preservation if other than specified (see 3.5.2).
(n) Level of packing required (see 5.2).
(o) Special marking if required (see 5.3).

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6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.3 and 4.4	DI-NDTI-80809	Test/inspection reports	----
4.3 and 4.4	DI-MISC-80678	Certification/data report	10.3.1 does not apply

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Selection instructions. For standardization and stocking purposes the following should be used as a guide for the acquisition of high pressure flasks:

(a) Flasks should be selected from the cubic capacities and working pressures specified in table III. When the desired capacity falls between the standard capacities listed, the next higher capacity should be used. When the system working pressure falls between those listed, the next higher working pressure flask should be used. Where it is impractical to conform to the standard capacities or working pressures specified in table III, specific approval of the cognizant activity should be obtained prior to the acquisition of any special flasks.

(b) The 1/2 cubic foot type SF (separator) and 1-1/2 cubic foot type GF flask should be acquired in the 5000 psi class only and should be used for all pressures up to and including 5000 psi. When a separator with a capacity larger than 1/2 cubic foot is needed, consideration should be given to the 1-1/2 cubic foot flask.

6.5 Subject term (key word) listing.

Compressed air
Gas storage
Helium
High-pressure gas
Impulse
Moisture separator

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6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Review activity:
DLA-GS

Preparing activity:
Navy - (SH)
(Project 8120-N387)

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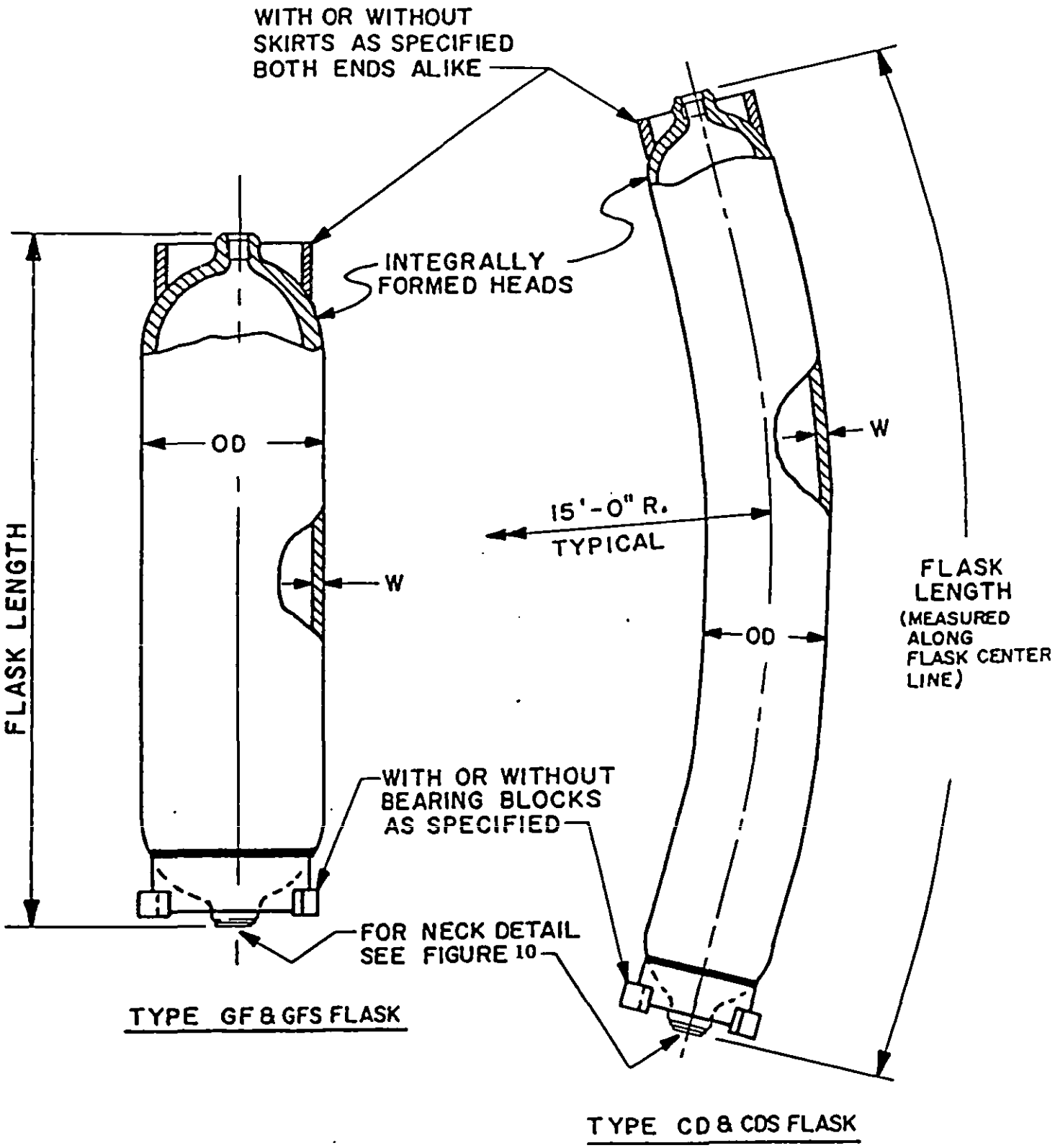


FIGURE 1. Flask types.

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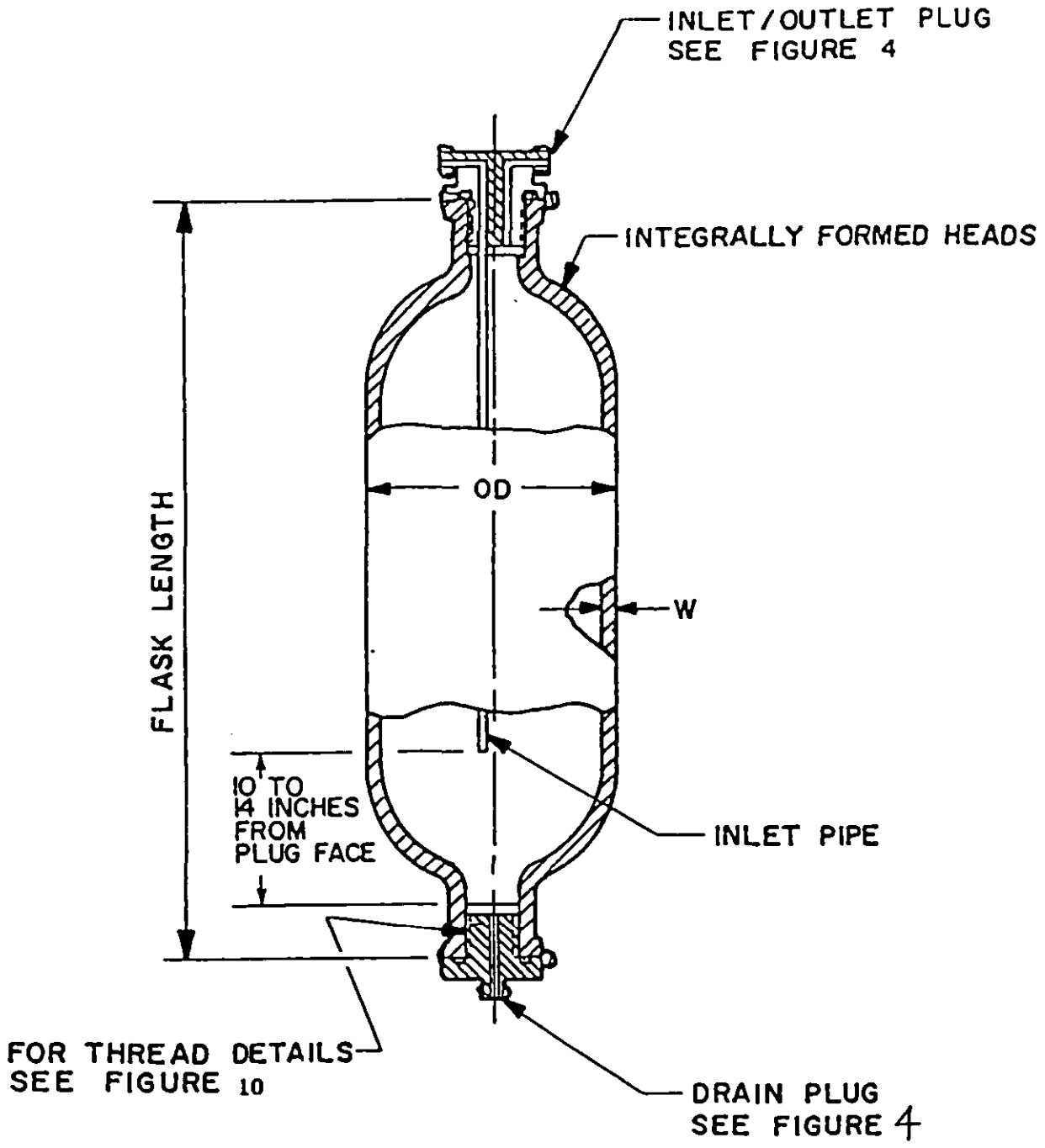
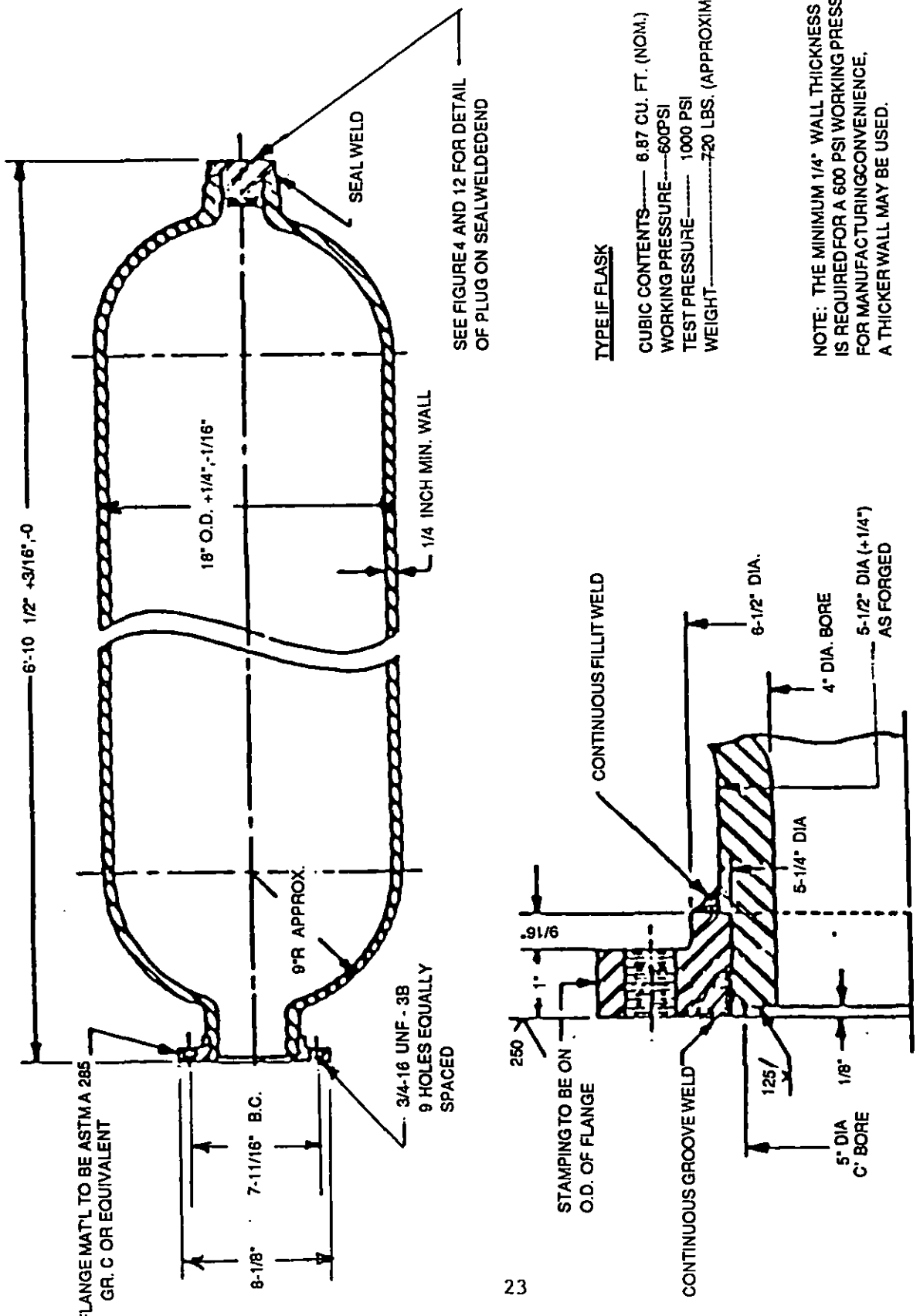


FIGURE 2. Type SF flask for vertical mounting only.

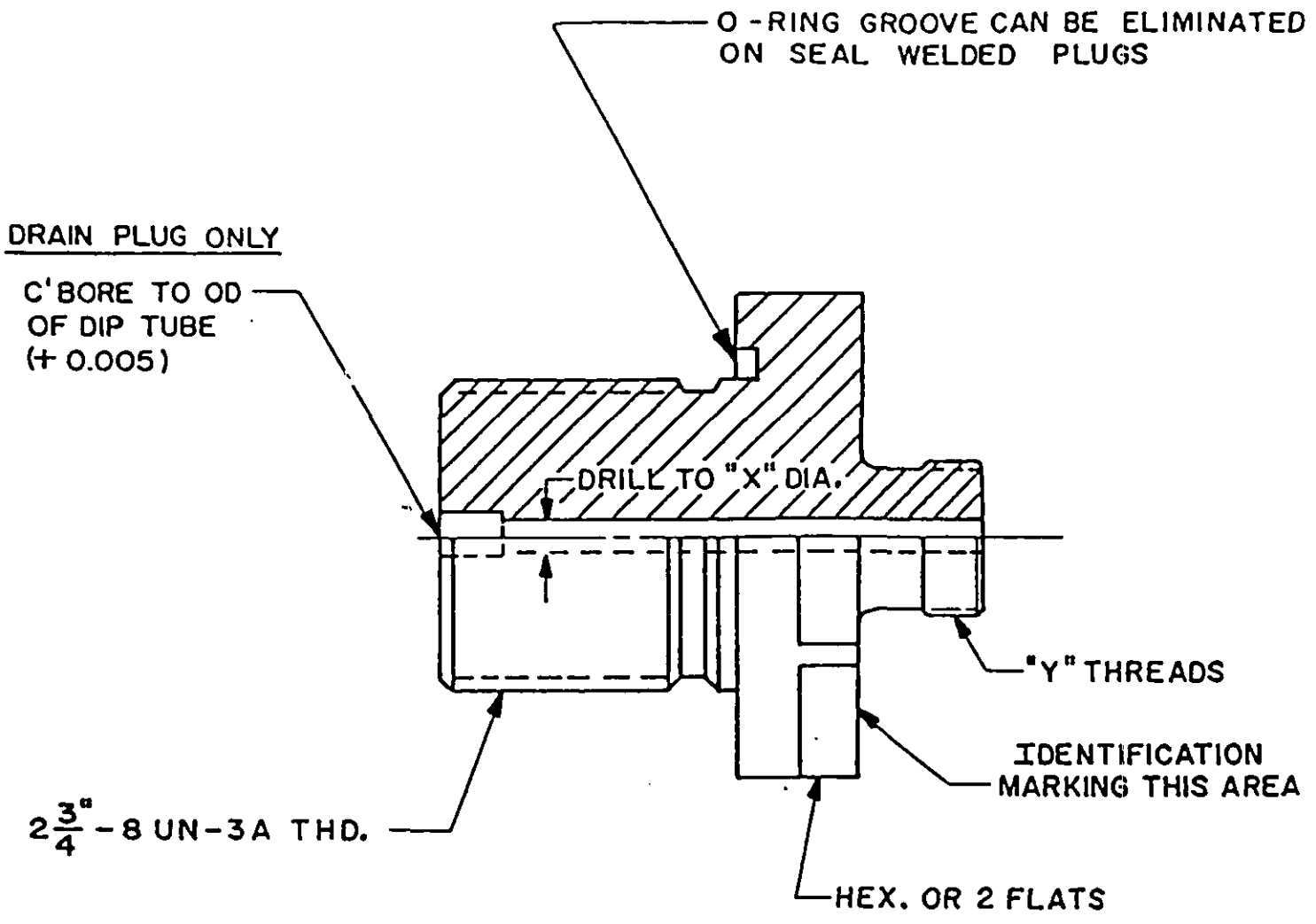
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DETAIL OF NECK AND FLANGE

FIGURE 3. IMPULSE FLASK

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Nominal pipe size	"X" Dia.	"Y" Threads	Remarks
1/4	5/16	1-3/16 - 12 UN-3A	Drain plug
3/8	7/16	1-3/8 - 12 UNF-3A	Inlet and outlet plug
1/2	1/2	1-3/4 - 12 UN-3A	Inlet and outlet plug
3/4	5/8	2 - 12 UN-3A	Inlet and outlet plug
1	13/16	2-5/16 - 12 UNS-3A	Inlet and outlet plug
1/2	13/16	1-3/4 - 12 UN-3A	Special
Solid	---	---	End closure plug no boss

FIGURE 4. Type STP inlet or outlet plugs for O-ring and welded assemblies.

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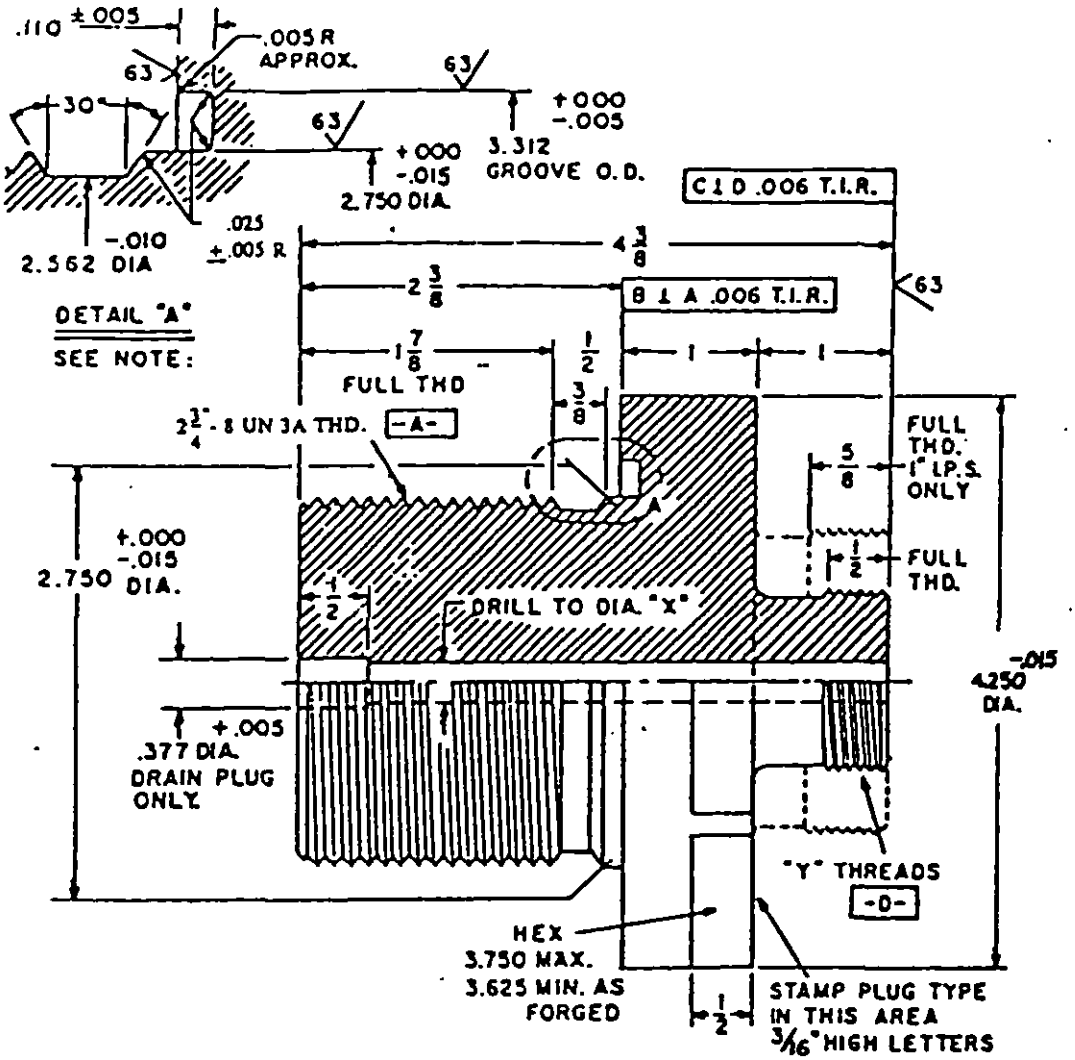
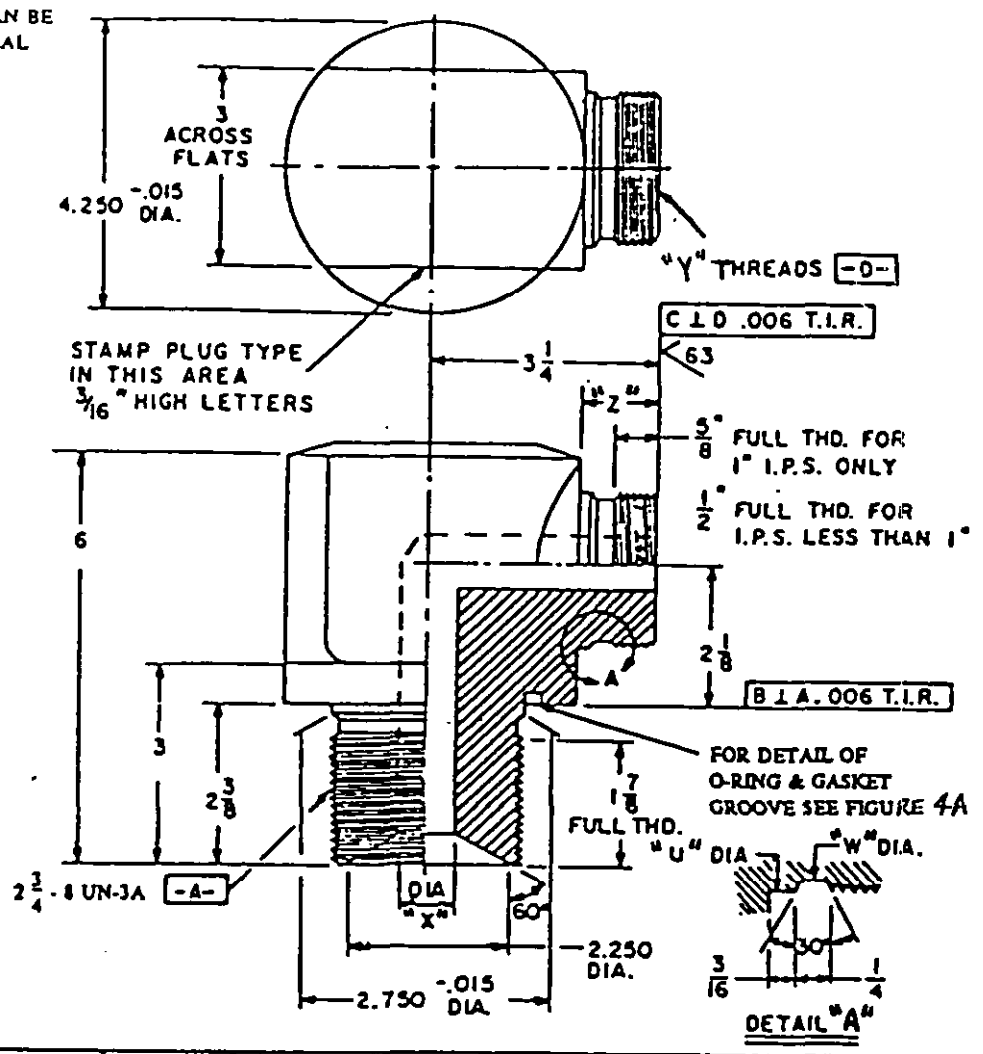


FIGURE 4A. Inlet or outlet plugs for O-ring and welded assemblies - Continued.

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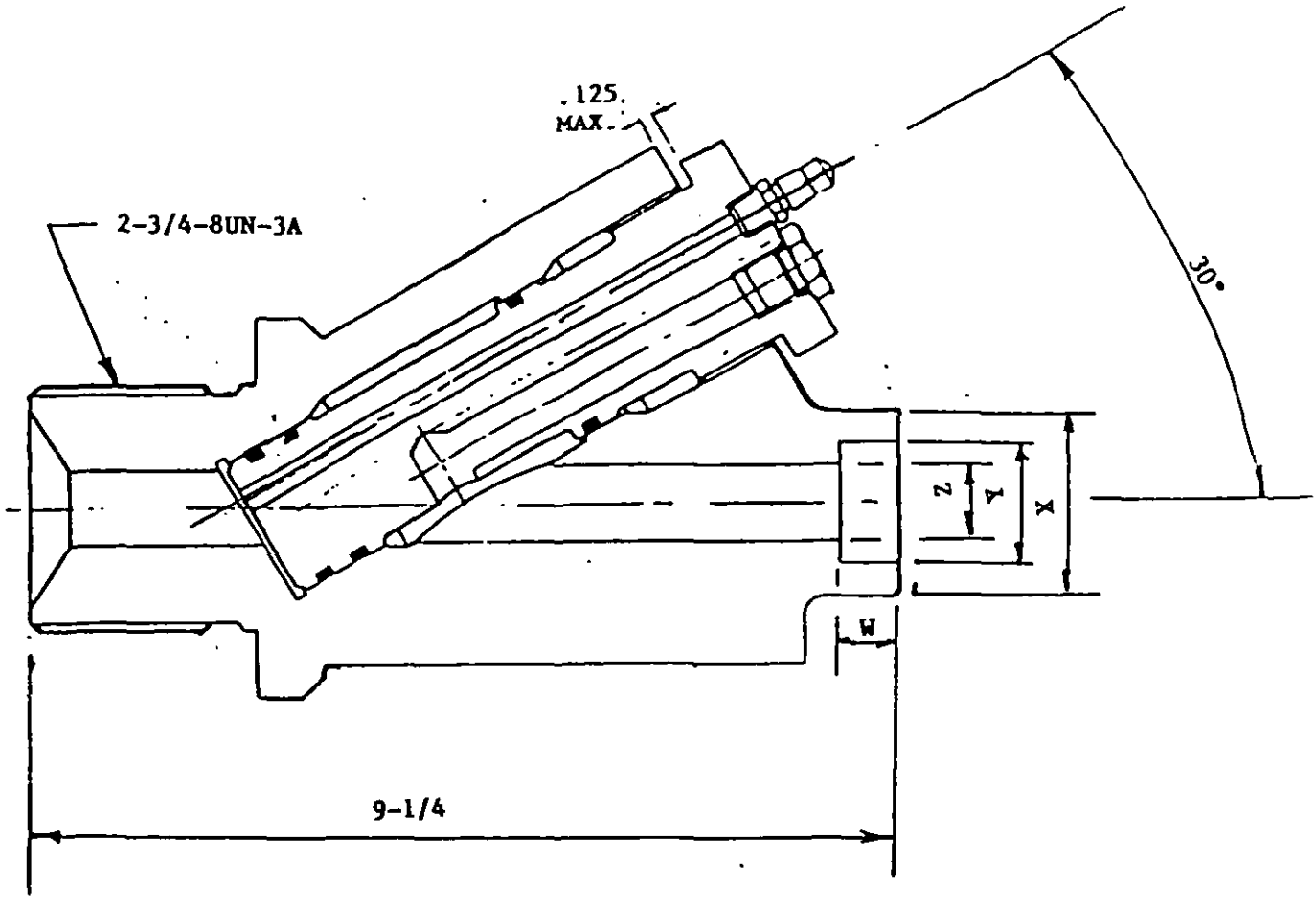
NOTE: O-RING GROOVE CAN BE ELIMINATED ON SEAL WELDED PLUGS.



NOMINAL PIPE SIZE	"X" DIA.	"Y" THREADS	"Z" +0.006	"U" DIA -0.010	"W" DIA -0.015	
1/4"	5/16"	1 3/16" -12UN - 3A	0.916	1.187	1.075	
3/8"	7/16"	1 5/8"-12 UNF -3A	.916	1.375	1.262	
1/2"	1/2"	1 3/4" -12 UN - 3A	.916	1.750	1.637	
3/4"	5/8"	2" -12 UN-3A	.916	2.000	1.887	
1"	13/16"	2 5/16"-12 UNS- 3A	1.041	2.3125	2.200	
1/2"	13/16"	1 3/4" -12 UN - 3A	.916	1.750	1.637	SPECIAL

FIGURE 5. Type RLP inlet or outlet plugs for O-ring and welded assemblies.

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Pipe	W Inch	X Inch	Y Inch	Z +/- 0.010 Inch
1 in. X 0.300 wall tube	0.62	2.34	1.323 1.328	0.631
1 in. X 0.250 wall pipe	0.62	2.06	1.323 1.328	0.815
0.5 in X 0.147 wall tube	0.50	1.28	0.847 0.852	0.548
0.25 in X 0.150 wall tube	0.50	1.05	0.547 0.552	0.198
0.25 in X 0.140 wall pipe	0.50	0.96	0.547 0.552	0.260

NOTE: All other details for the "Y" plug shall be as specified by the procuring activity.

FIGURE 6. Type "Y" inlet/outlet plug.

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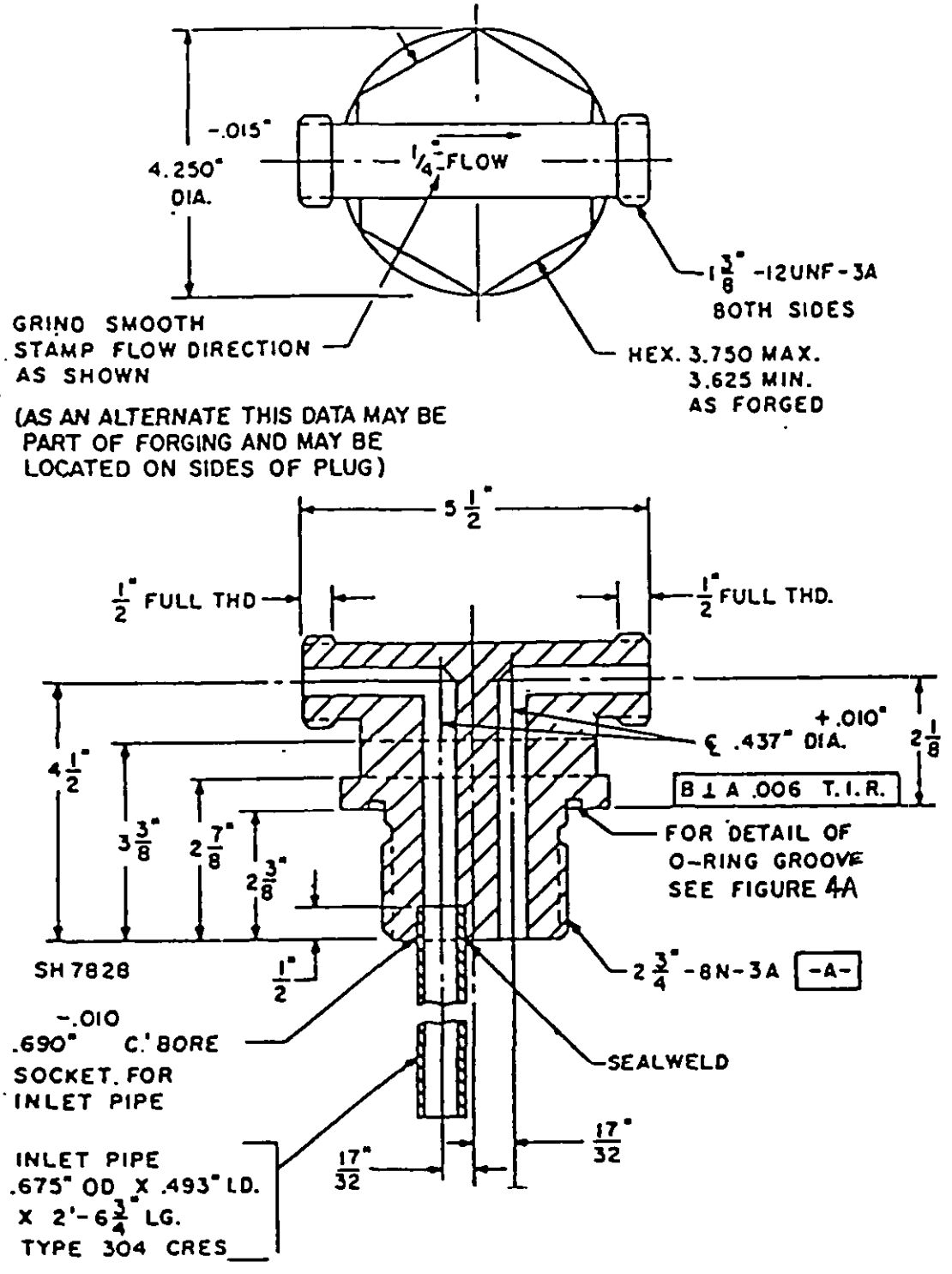


FIGURE 7. Type SFP inlet and outlet plug for type SF (separator) flask.

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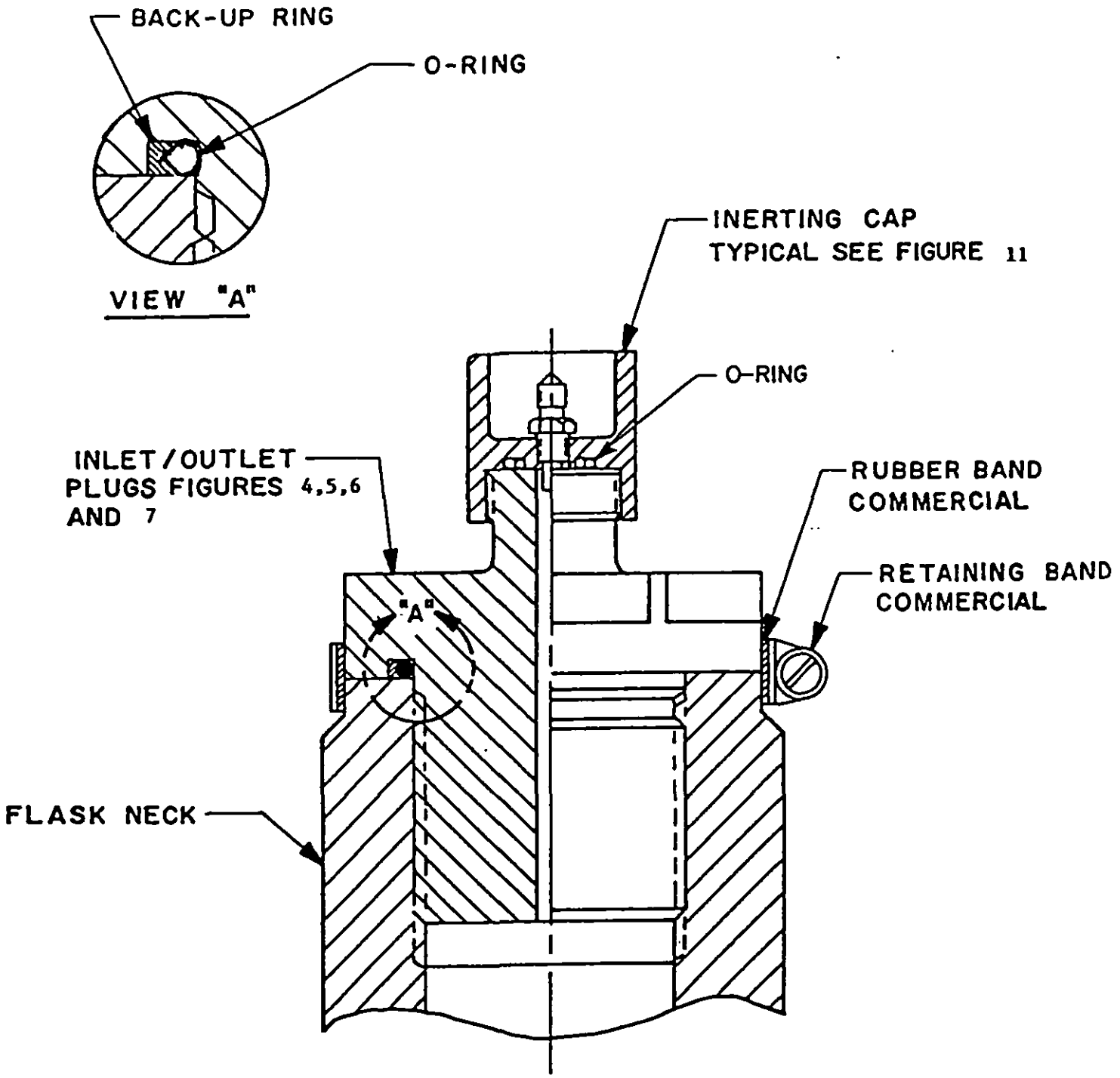


FIGURE 8. Assembly inlet or outlet plugs O-ring - nonwelded.

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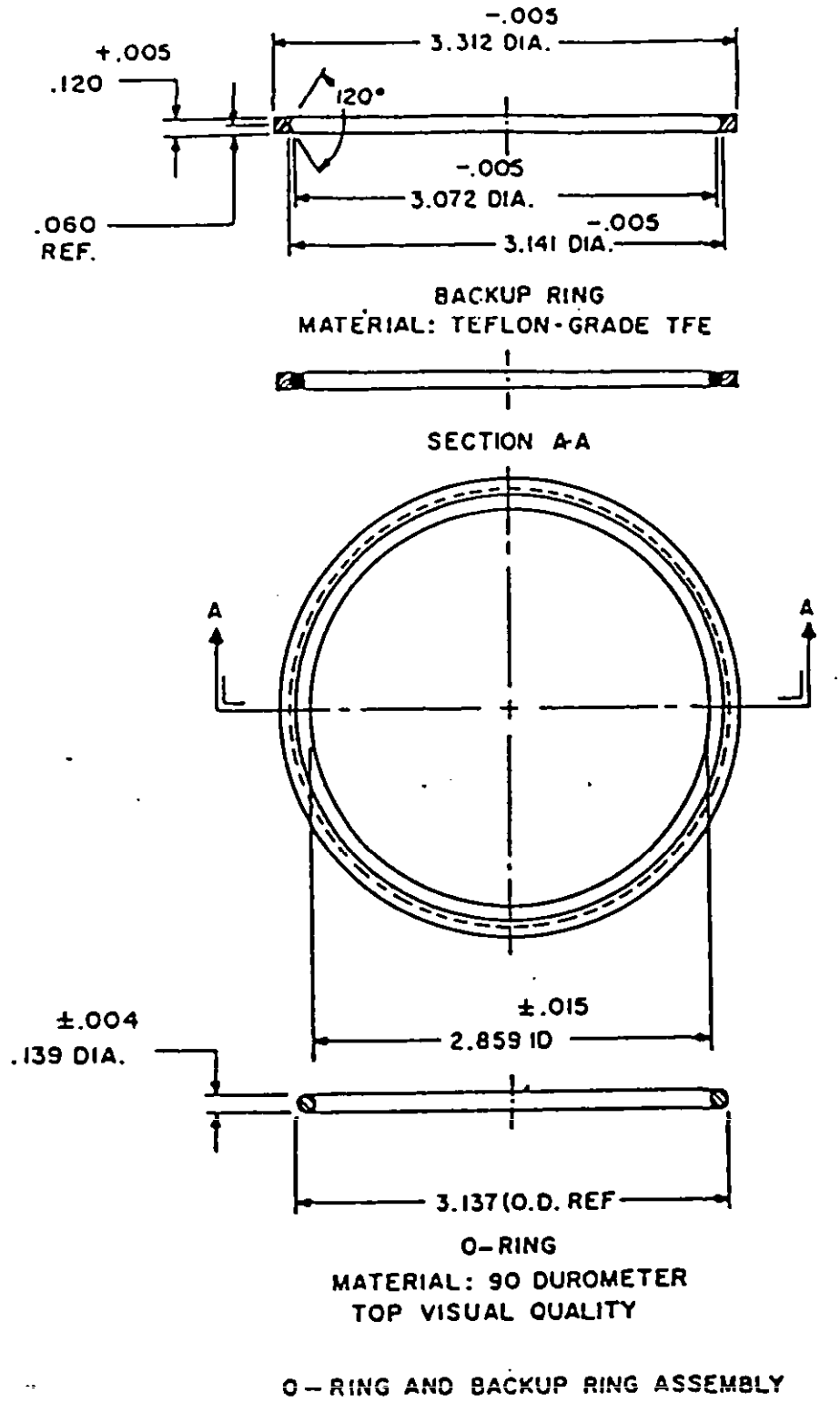


FIGURE 9. Assembly inlet or outlet plug O-ring - nonwelded.

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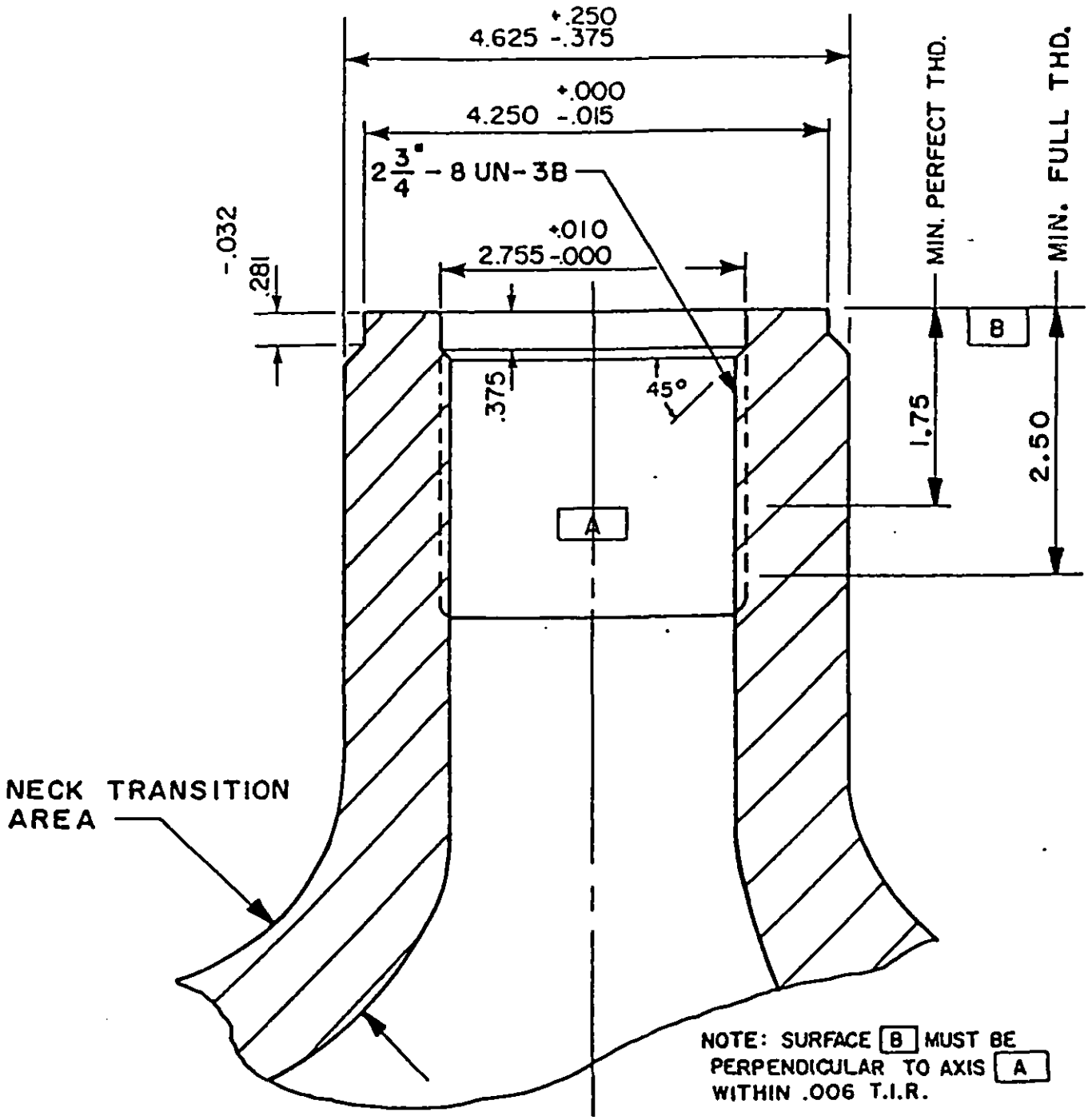
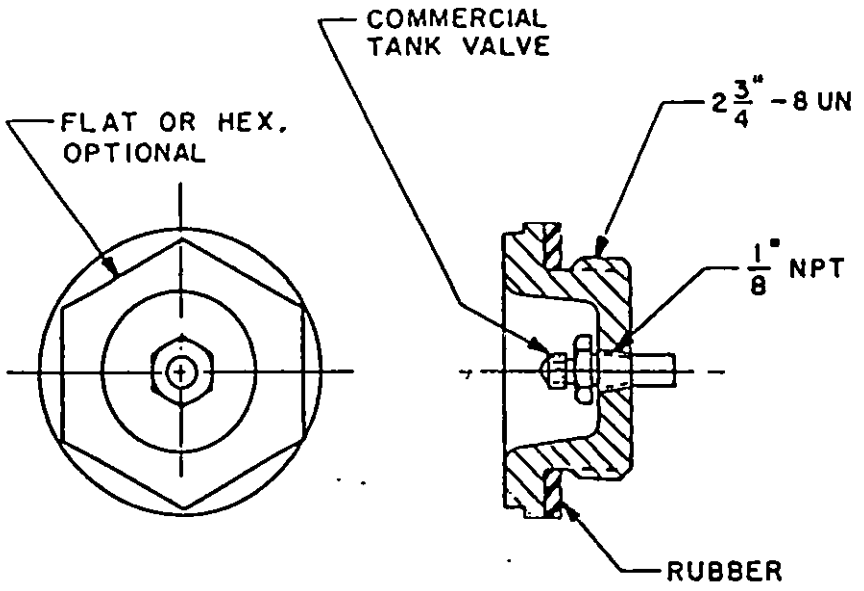
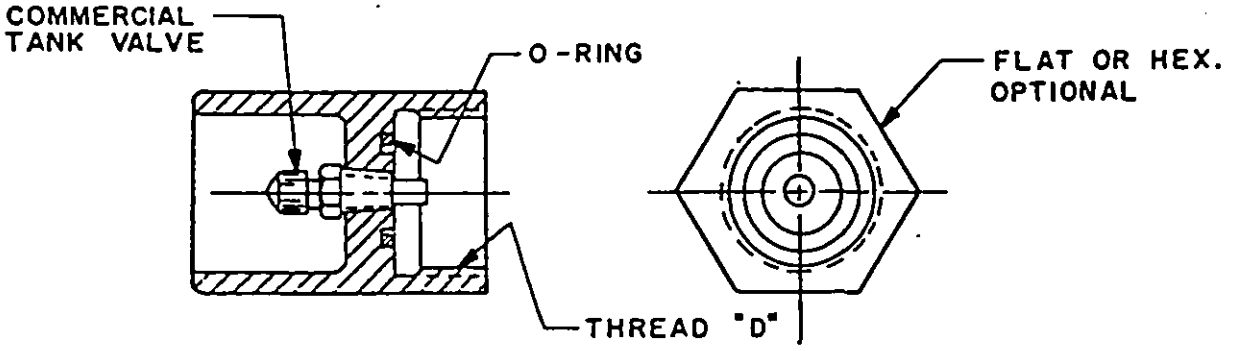


FIGURE 10. Neck thread details.

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INERTING CAP - TYPE A
WITHOUT TANK VALVE-TYPE B - SOLID



INERTING CAP — TYPE C
WITHOUT TANK VALVE -TYPE D - SOLID

Iron pipe size	"D" threads
1/4	1-3/16 - 12 UN
3/8	1-3/8 - 12 UNF
1/2	1-3/4 - 12 UN
3/4	2 - 12 UN
1	2-5/16 - 12 UNS

NOTES:
1. Cap must withstand a pressure of 100 psi.
2. All surfaces must be corrosion resistant.

FIGURE 11. Inerting caps.

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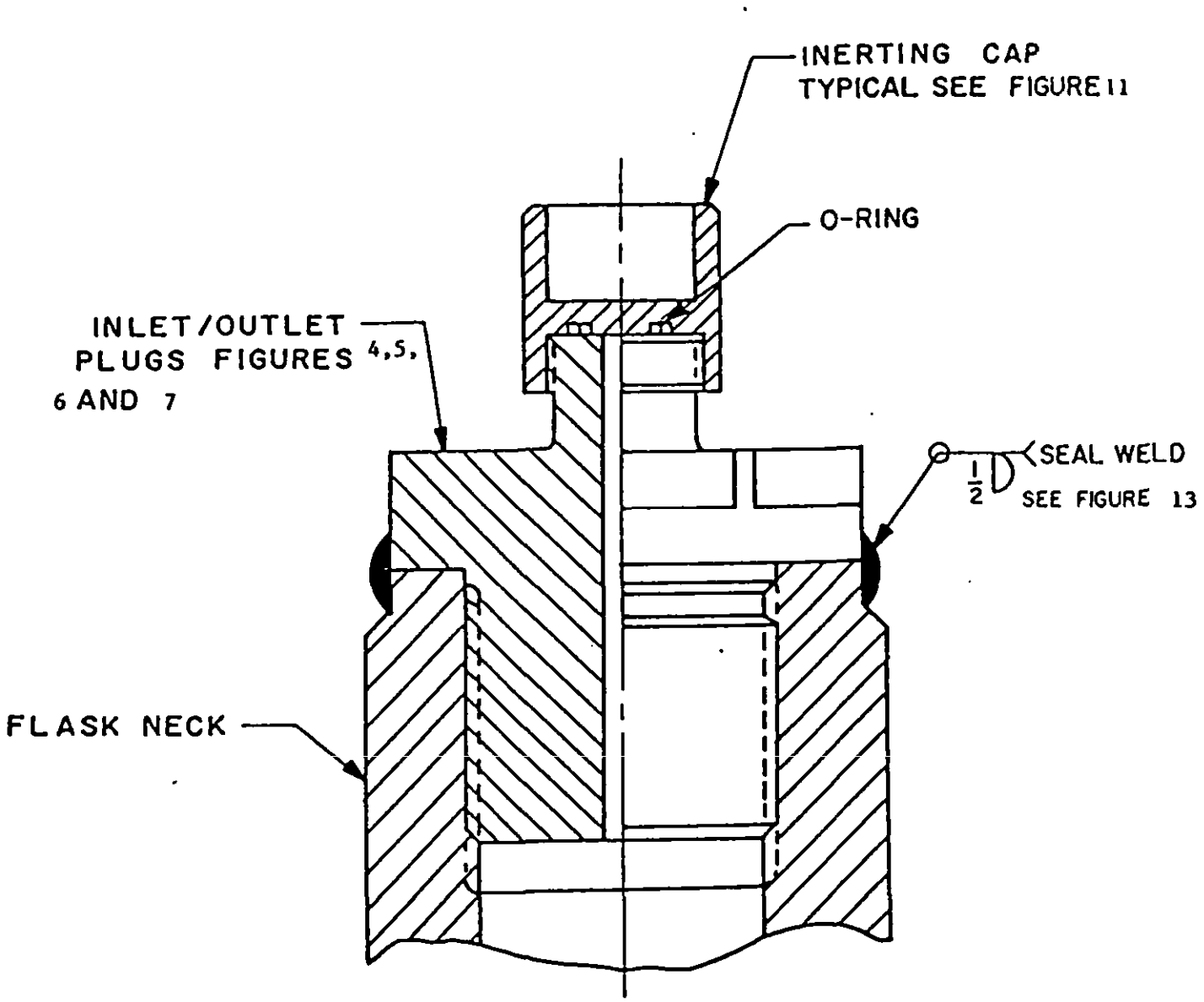
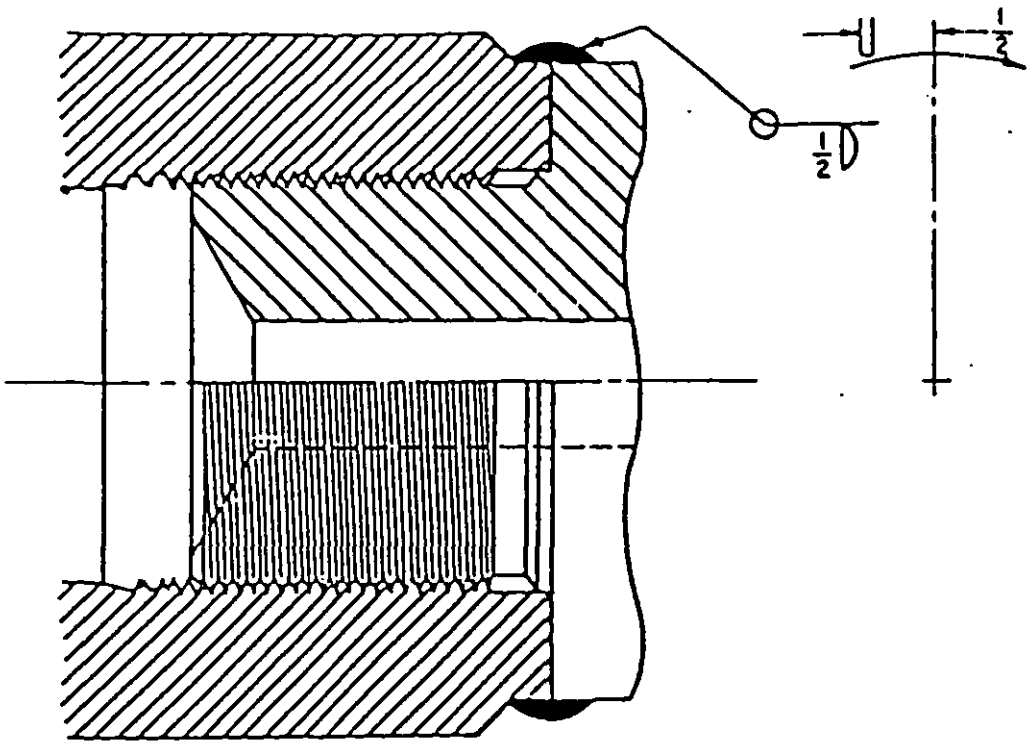


FIGURE 12. Assembly inlet or outlet plugs seal welded.

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TYPE OF WELD-SEAL WELD



SEAL WELD PROCEDURE
TORQUE PLUG TO 800 ± 200 FOOT-POUNDS

PREPARATION - CLEAN AREA TO BE SEAL WELDED INCLUDING ABUTTING FACES OF CYLINDER NECK AND PLUG.
POSITION - HORIZONTAL ROLL WITH ELECTRODE 1/2 INCH AHEAD OF TOP CENTER AS ILLUSTRATED.
PREHEAT - 100 TO 200°F MAXIMUM.
ELECTRODE - AWS ASTM E 309-16 PREFERRED, 1/8 INCH DIAMETER, USING DC REVERSE POLARITY CURRENT AT 24-26 VOLTS AND 80-100 AMPERES.
NO. OF BEADS - TWO, WITH ALL STARTS AND STOPS STAGGERED. CLEAN SLAG WITH BLUNT TOOL AND STAINLESS STEEL WIRE BRUSH BEFORE CONTINUING A WELD.
ARC TRAVEL SPEED - 1ST LAYER AT 9 TO 11 INCHES PER MINUTE.
2ND LAYER AT 8 TO 12 INCHES.
HEAT TREATMENT - NONE.

NOTES:
THE ELECTRODE AND PROCEDURE RECOMMENDED WITH YIELD A CORROSION RESISTANT, LOW FLAT WELD WITH MINIMUM PENETRATION, FOR EASE OF LATER REMOVAL BY GRINDING. TWO LAYERS ARE USED TO INSURE AGAINST POROSITY LEAKAGE. THE "AHEAD OF CENTER" POSITION SHOULD NOT BE EXCEEDED AS MOLTEN METAL AND SLAG MAY RUN AHEAD OF ARC AND CAUSE A FAULTY WELD.

FIGURE 13. Seal welded typical.

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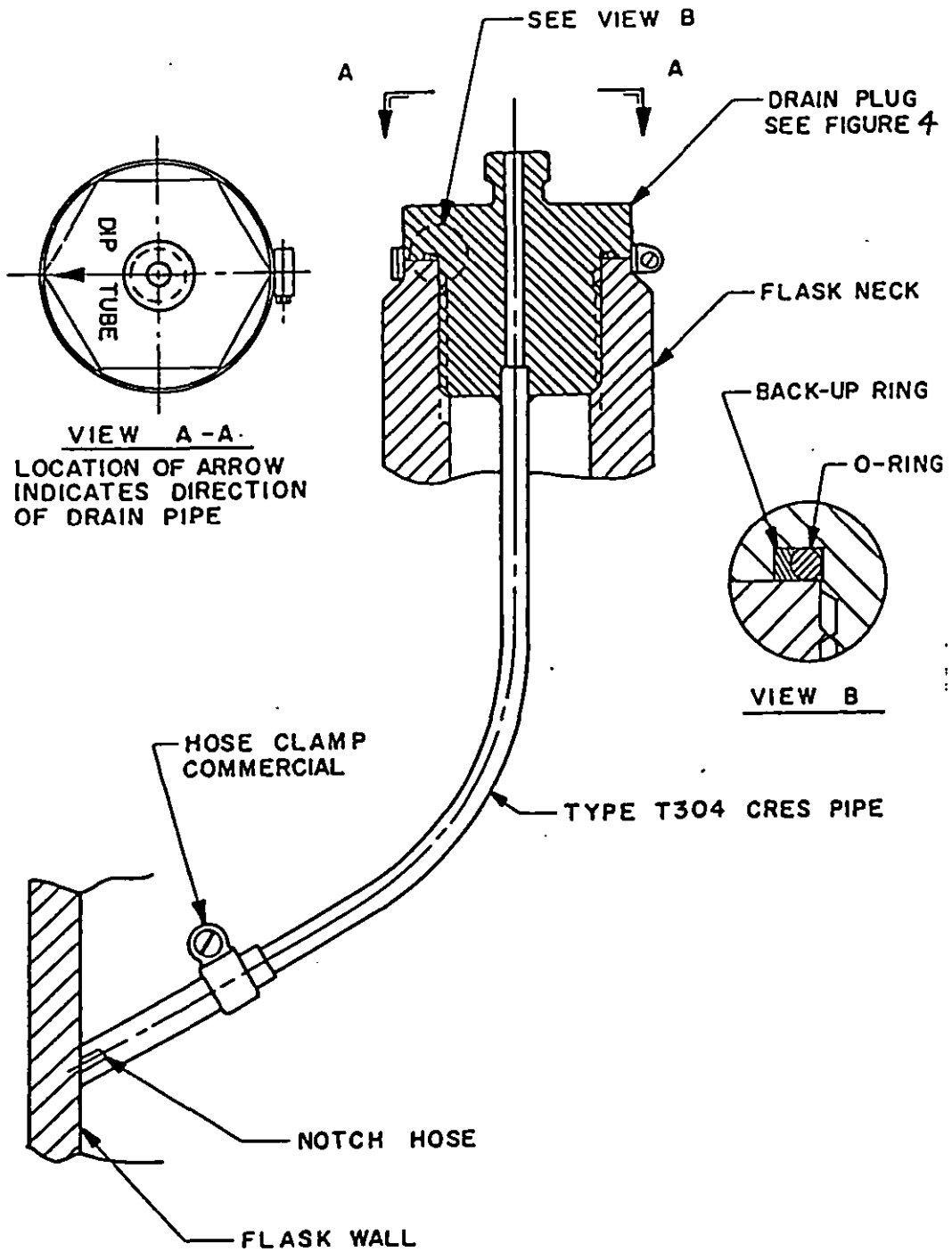


FIGURE 14. Internal drain piping assembly for flasks installed horizontally.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-F-22606C	920811

3. DOCUMENT TITLE
FLASK COMPRESSED GAS AND END PLUGS FOR AIR, OXYGEN AND NITROGEN

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
TECHNICAL POINT OF CONTACT: Mr. Carl Banks, SEA 05Y23	(1) Commercial 703-602-0367	(2) AUTOVON 332-0367
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command SEA 05Q42, 2531 National Center Bldg 3 Washington, DC 20362-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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